

Hospital Capacity Planning and Optimisation

Optimising weekly procedure-room configurations for a GI unit using mixed-integer linear programming.

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Hospital procedure-room optimisation case study

This portfolio case study reframes an academic optimisation assignment as a business-facing healthcare operations report. The analysis focuses on a gastrointestinal (GI) unit that must configure procedure rooms each week to meet diagnostic and therapeutic demand at minimum cost while remaining within clinician capacity.

10 procedure rooms modelled	26 weeks planning horizon	2 procedure streams: diagnostic and therapeutic	2,265,600 minimum total cost units
87,138 average weekly cost units	7 average rooms used per week	8 total room setup activations	0.00883% MILP optimality gap

Headline result

The optimisation model produced a 26-week room configuration plan that met all diagnostic and therapeutic demand, respected weekly clinician limits, and minimised total operating cost within solver tolerance.

// BUSINESS QUESTION

What decision needed to be made?

The GI unit faced a semiannual capacity planning problem. Demand for diagnostic and therapeutic procedures varied week by week, rooms had different capacities and costs, and clinician availability was limited. Management needed a clear weekly plan for which rooms to activate, how to configure them, and how to avoid unnecessary setup and allocation costs.

Business question

How should a hospital configure procedure rooms each week to meet diagnostic and therapeutic demand at minimum total cost, without exceeding clinician availability?

From capacity planning problem to optimisation model

The planning model used weekly demand, clinician availability, room capacity, allocation cost and setup cost data. Rooms were grouped into small, medium and large profiles, with different diagnostic and therapeutic capacities. The optimisation model was formulated as a mixed-integer linear programming problem and solved in R.

Inputs used in the model

Input	Decision relevance
Room capacity	Small, medium and large rooms with different weekly diagnostic and therapeutic capacities.
Weekly demand	Diagnostic and therapeutic procedure-hour requirements for each of the 26 weeks.
Clinician availability	Weekly limits on total procedure hours that can be staffed.
Cost parameters	Room allocation costs and setup costs for bringing rooms into active GI use.
Configuration rules	Each room can be diagnostic, therapeutic or unavailable in any week.

Optimisation workflow

- Defined binary room-configuration decisions for each room and week.
- Defined continuous variables for diagnostic and therapeutic hours scheduled in each room-week.
- Minimised total allocation and setup cost across the 26-week planning horizon.
- Enforced demand fulfilment, room capacity, clinician availability and setup-activation constraints.
- Validated the final solution using solver output, weekly cost totals and feasibility checks.

What the optimisation revealed

1. Demand can be met without using every room

The recommended solution uses an average of seven rooms per week, with only selected higher-demand weeks requiring eight rooms.

2. Setup costs are concentrated

Most setup costs occur at the start of the horizon, with an additional setup in week 9 when one larger room is activated for higher demand.

3. Small and medium rooms carry most of the operating plan

Large rooms are used selectively, which helps meet demand without carrying unnecessary high-cost capacity every week.

4. Feasibility is as important as cost

The model does not simply reduce cost. It also confirms every week meets diagnostic and therapeutic demand and stays within clinician limits.

Recommended operating plan

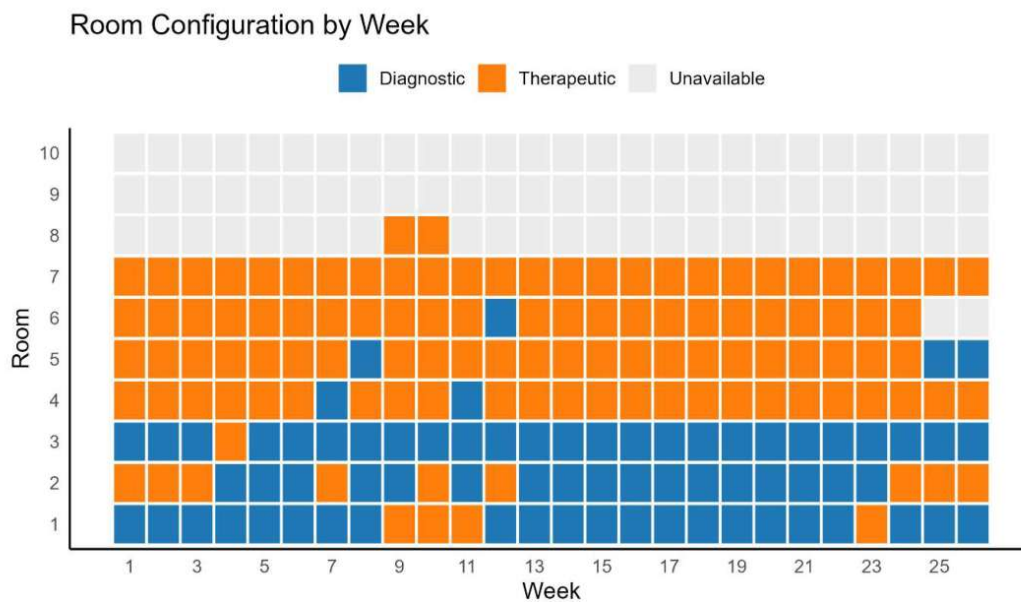


Figure 1. Room configuration by week

Business takeaway

The weekly configuration plan shows which rooms should be diagnostic, therapeutic or unavailable. This converts the optimisation result into a usable operating schedule for managers.

Capacity plan validation

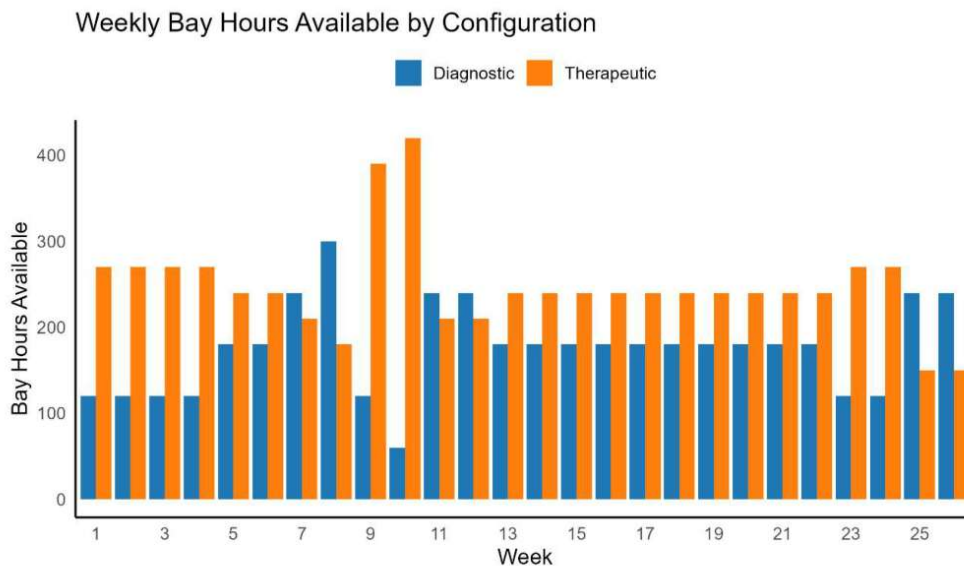


Figure 2 Weekly Bay Hours Available by Configuration

Figure 2. Weekly bay hours available by configuration

Business takeaway

The plan creates sufficient diagnostic and therapeutic capacity across the full horizon, including higher-demand weeks where capacity requirements increase.

Capacity use and cost behaviour

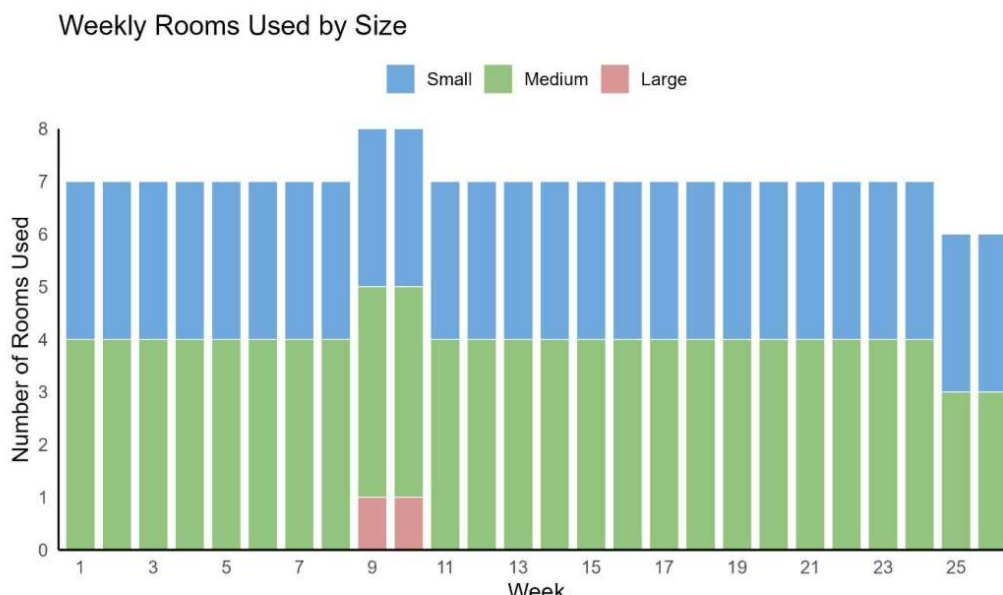


Figure 3. Weekly rooms used by size

Business takeaway

Most required capacity is delivered through small and medium rooms, while large rooms are only introduced where the demand profile justifies the additional capacity and cost.

Figure 3 Weekly Rooms Used by Size

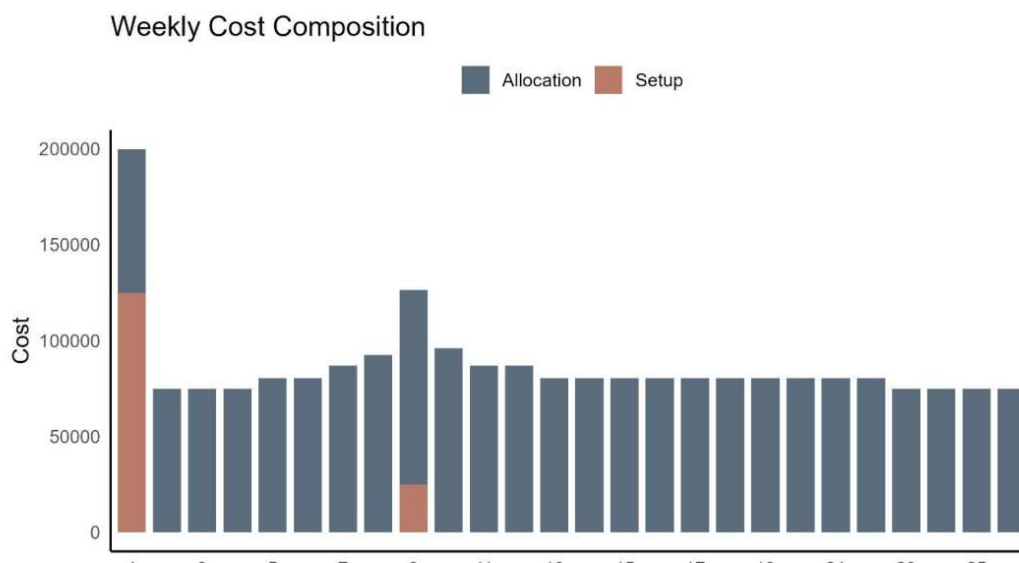


Figure 4. Weekly cost composition

Business takeaway

Weekly costs are primarily allocation costs after the initial setup. Setup cost appears strongly in week 1 and again in week 9, when an additional large room is activated.

// OPTIMISATION PERFORMANCE

Solver output and feasibility validation

The model was solved in R using a MILP workflow. The solver reported a best feasible objective value of 2,265,600 and a lower bound of 2,265,400. This produced a relative optimality gap of 0.00883%, which sits below the default tolerance threshold used in the assignment.

Validation metric	Result
Best feasible objective	2,265,600
Best lower bound	2,265,400
Relative MILP gap	0.00883%
Total allocation cost	2,115,600
Total setup cost	150,000
Demand met	Yes, all 26 weeks
Clinician capacity respected	Yes, all 26 weeks
Average rooms used	7 per week

Business interpretation

The result is not just a mathematical optimum. It is a week-by-week operating plan that balances demand fulfilment, room capacity, staffing feasibility and cost control.

How the analysis supports managers

The optimisation model creates practical decision value by translating a complex capacity planning problem into a clear weekly schedule. It helps managers understand when rooms should be active, which configuration they should use, when setup costs are triggered and whether the final schedule is feasible.

- Supports semiannual planning with a structured 26-week room configuration plan.
- Prevents unnecessary room activation by explicitly modelling setup costs.
- Keeps the plan operationally realistic by enforcing weekly clinician availability.
- Provides a validation trail showing total cost, demand fulfilment and capacity feasibility.
- Creates a repeatable decision-support approach for future planning cycles.

// RECOMMENDATIONS

Business recommendations from the optimisation

1. Use the MILP plan as the baseline room configuration schedule

The recommended 26-week plan should be treated as the baseline operating plan because it meets all procedure demand while minimising total cost within solver tolerance.

2. Keep large rooms reserved for higher-demand periods

Large rooms provide more capacity but are more expensive. The solution shows they should only be activated when the weekly demand profile justifies the additional capacity.

3. Track setup triggers carefully

Setup costs materially affect total cost. Management should avoid switching rooms off and on unnecessarily unless demand changes make it worthwhile.

4. Monitor clinician capacity as a binding operational constraint

The optimisation confirms staffing feasibility week by week. If clinician availability changes, the model should be rerun because room capacity alone does not guarantee a feasible plan.

5. Use the model as a scenario-planning tool

The same model can be reused to test demand increases, staff shortages, room closures, cost changes or alternative service mixes.

What should be considered before implementation

Limitations

- The model is deterministic and assumes known weekly demand and clinician availability.
- It does not explicitly model cancellations, emergency cases, equipment downtime or day-to-day operational disruption.
- The cost structure is based on the assignment parameters and may need recalibration for a real hospital setting.
- The model provides a tactical weekly plan rather than a real-time scheduling system.

Next steps

- Create demand scenarios for low, expected and high procedure volumes.
- Add sensitivity analysis around clinician availability and room setup costs.
- Convert outputs into a dashboard for managers to compare weekly demand, capacity and cost.
- Test whether more flexible room switching could improve resilience under uncertain demand.
- Extend the model to include waiting-list penalties or service-priority weights.

// PORTFOLIO REFLECTION

What this project demonstrates

This project demonstrates how optimisation can support practical healthcare operations decisions. The technical model is important, but the key portfolio value is the translation of an MILP solution into a clear managerial plan: which rooms to use, when to activate them, what cost is incurred, and whether the plan is feasible.

The project strengthened my understanding of how business analytics can move beyond reporting into prescriptive decision support. It shows my ability to formulate a real operational problem, code an optimisation model in R, validate the solution and explain the result in business terms.

Tools and methods used

Tool or method	Role in the project
R	Model implementation, data setup, solver execution and output validation.
ompr	Mathematical optimisation modelling in R.
HiGHS solver	MILP solution and optimality gap reporting.
MILP formulation	Binary room configuration variables, continuous scheduled-hours variables and setup decision variables.
Healthcare operations analytics	Capacity planning, cost minimisation, room utilisation and staffing feasibility.

Source note: This employer-facing case study is based on the submitted ITAO 7104 Data-Driven Decision-Making assignment and has been rewritten into a portfolio business report format.